

Project Sonar: Short-Horizon Equity Direction Prediction with Multi-Scale Deep Learning

Team name: Stock Project | Team member: Scottie Dunnavaan | Deliverable: Problem definition & project proposal (PS2)

1. Problem Statement

Retail traders compete against institutions that run algorithmic and high-frequency strategies on proprietary data and infrastructure. Manual chart reading cannot keep pace with the volume of live order-book updates and is prone to emotional bias. This project targets one narrow, measurable version of that gap: **short-horizon (1–5 minute, intraday) price-direction prediction**. Given recent limit order book (LOB) and price-history data for a stock, a model estimates the probability that the mid-price will move *Up*, *Neutral*, or *Down*, and those probabilities are turned into entry/exit signals under strict risk limits.

Why it matters: whether deep learning can extract a usable signal from data an individual can actually access, and how much of the published LOB results survive outside curated benchmarks, are open and practical questions. Because financial data is noisy and non-stationary, the project emphasizes rigorous, leakage-free evaluation over headline accuracy; a well-evaluated negative or marginal result is still a valid outcome.

2. Motivation & Objectives

Motivation. I have a long-standing interest in the mathematics of markets, inspired by quantitative firms such as Renaissance Technologies that trade systematically using mathematical and statistical-learning methods. This is a personal, hobby and investment-tool project, and a portfolio project: I want to learn the foundations of algorithmic trading and deep learning by building an end-to-end system (paper trading only in this course). **Why deep learning:** LOB data is high-dimensional (prices and volumes across many levels), sequential, and noisy. Hand-crafted indicators discard much of that structure, while convolutional and recurrent/attention models can learn spatial and temporal patterns directly. DeepLOB [1] showed this on benchmark data; this project tests how far it carries to live US equity data.

Objectives (task type: 3-class classification of price direction):

- **O1 (core):** Reproduce DeepLOB on the FI-2010 benchmark to validate the pipeline.
- **O2 (core):** Build classical baselines (Random Forest, XGBoost) on technical and order-book features.
- **O3 (core):** Train deep models on self-recorded US equity data for 1–5 minute direction.
- **O4 (core):** Evaluate with chronological splits and transaction costs.
- **O5 (stretch):** Fuse micro (LOB) and macro (price-history) models, add a risk-managed decision layer, and run paper trading.

Aspiration: a win rate of 70% or higher on recommended trades. Because win rate alone does not determine profitability, this is a stretch goal; the formal success criteria are in Section 5.

3. Dataset Description

The data plan has two phases: a public benchmark (FI-2010) to validate the pipeline, then self-recorded live US data for the main experiments, supported by Level 1 and daily data. Sources are listed in the table below.

- **Inputs and preprocessing:** sliding windows of recent LOB snapshots (e.g., 100 time steps × 40 features: 10 price/volume levels per side) and OHLCV windows for the sequence model. Rolling z-score normalization uses past data only; log returns give stationarity.
- **Labels:** Up/Neutral/Down from the thresholded change in (smoothed) future mid-price over a 1–5 minute horizon. Thresholds are set on training data for reasonable class balance, with a class-weighted loss for any remaining imbalance.
- **Splits:** strictly chronological train/validation/test (plus walk-forward for backtests), no shuffling, with a gap between splits so overlapping windows cannot leak information.

- **Limitation:** FI-2010 is small, dates from 2010, is from Nordic markets, and uses event-count horizons. It is used for validation only; its results are not assumed to transfer to live US data.

Data sources

Source	Contents, size, format	Cost / constraints	Role
FI-2010 benchmark [2]	5 Nasdaq Nordic stocks, 10 trading days (June 2010), ~4 million samples; top 10 bid/ask price and volume levels; Up/Stationary/Down labels at horizons of 10–100 events; text files, pre-normalized.	Free.	Phase 1: reproduce DeepLOB and validate the pipeline.
IBKR TWS API (live Level 2) [5]	Self-recorded depth snapshots (top 5–10 levels) for a small, liquidity-screened universe of US small-to-mid-cap stocks (target 5–10 tickers) during market hours; stored as Parquet.	Needs an IBKR account and market-data subscription; simultaneous depth streams are limited; no historical depth, so recording starts in Week 1.	Phase 2: primary data for own-data models (target 6–8+ weeks of recordings).
Massive (formerly Polygon.io) [6]	Level 1 data: trades, NBBO quotes, and 1-minute bars via REST and WebSocket. No Level 2 depth for US stocks.	Free tier; paid tiers for real-time.	L1 features, historical bars, cross-check of recorded data.
Yahoo Finance (yfinance) [8]	Daily OHLCV history; intraday history limited to a short recent window.	Free; rate-limited (caching and exponential backoff).	Daily/regime features and baselines.

4. Proposed Methodology

- **Baselines (core):** majority-class predictor; Random Forest and XGBoost [4] on RSI, MACD, Bollinger Bands, VWAP, lagged returns, bid-ask spread, and order-book imbalance.
- **Deep learning (core, PyTorch):** DeepLOB [1] (convolutions + Inception module + LSTM) on LOB windows; an LSTM or Temporal Fusion Transformer [3] on 1-minute OHLCV sequences. Softmax output over Up/Neutral/Down.
- **Transfer learning (core experiment):** initialize DeepLOB from FI-2010-pretrained weights, fine-tune on recorded US data, and compare against training from scratch.
- **Multi-scale fusion (stretch):** concatenate embeddings from the LOB (micro) and price-history (macro) branches into a joint classifier head. Optional second modality: news-sentiment embeddings (e.g., FinBERT).
- **Risk-managed decision layer (stretch):** trade only above a probability threshold; capped fractional-Kelly sizing; ATR-based stop-loss/take-profit; circuit breakers at 2% daily and 5% cumulative drawdown. Backtest and paper trading only; no real capital.
- **Tooling:** Python, PyTorch, scikit-learn/XGBoost, pandas/Parquet, and ib_async [7] for the IBKR API.

5. Evaluation Plan

- **ML metrics:** per-class precision, recall, and F1, with **macro-F1 as the primary metric**; confusion matrices; accuracy is always read against the majority-class baseline because “Neutral” can dominate at short horizons.
- **Financial metrics** (backtests include spread, commissions, and slippage assumptions): win rate, average win/loss, expected value per trade, $EV = (Win\ Rate \times Avg\ Win) - (Loss\ Rate \times Avg\ Loss)$, Sharpe and Sortino ratios, and maximum drawdown.
- **Comparisons:** majority baseline → Random Forest/XGBoost → LSTM/TFT → DeepLOB → (stretch) fused model; trading results versus Buy & Hold and a dual moving-average crossover.
- **Success criteria:** (1) the FI-2010 reproduction is comparable to published DeepLOB results; (2) on held-out chronological test data from my own recordings, macro-F1 exceeds the best baseline; (3) EV per trade is positive after costs in backtests. A 70%+ win rate remains the stretch target.

6. Expected Outcomes & Risks

Expected outcomes: DeepLOB should reproduce well on FI-2010; smaller and less certain gains are expected on live US data, and I will report them honestly. Deliverables: a reproducible data pipeline, a baseline-versus-deep comparison, a cost-aware backtest, and (stretch) a paper-trading demo.

Key risks and mitigations

Risk	Mitigation
Noise / overfitting (low signal-to-noise ratio)	Regularization, early stopping, walk-forward validation.
Benchmark-to-live gap (FI-2010 vs. live US data)	Own-data phase; FI-2010 used for validation only.
Data access (L2 needs a paid subscription; no historical depth from the sources above)	Start recording in Week 1; fall back to L1 (NBBO) plus bars if L2 coverage is insufficient.
Concept drift	Periodic retraining and walk-forward testing.
Execution latency and costs	Cost-aware backtests; paper trading only.
Scope (solo, 11 weeks)	Core/stretch split protects the schedule.

7. Timeline & Milestones (11 weeks)

Weeks	Dates	Milestones
1-2	Sep 20 – Oct 3	Set up repo and API access (IBKR, Massive, yfinance); start LOB recording ; download FI-2010.
3-4	Oct 4 – Oct 17	Preprocessing, labeling, splits; Random Forest/XGBoost baselines.
5-7	Oct 18 – Nov 7	Reproduce DeepLOB on FI-2010; train on recorded data; LSTM/TFT sequence model; transfer-learning experiment.
8-9	Nov 8 – Nov 21	Cost-aware backtesting and model comparison; stretch: multi-scale fusion.
10	Nov 22 – Nov 28	Stretch: risk layer and paper-trading validation.
11	Nov 29 – Dec 5	Final experiments, figures, final report and presentation.

Dates will be aligned with the milestone deadlines posted on Canvas.

8. References

- [1] Zhang, Z., Zohren, S., & Roberts, S. (2019). DeepLOB: Deep Convolutional Neural Networks for Limit Order Books. *IEEE Transactions on Signal Processing*, 67(11), 3001–3012.
- [2] Ntakaris, A., Magris, M., Kannianen, J., Gabbouj, M., & Iosifidis, A. (2018). Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods. *Journal of Forecasting*. doi:10.1002/for.2543
- [3] Lim, B., Arik, S. Ö., Loeff, N., & Pfister, T. (2021). Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting. *International Journal of Forecasting*, 37(4), 1748–1764.
- [4] Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of KDD 2016*, 785–794.
- [5] Interactive Brokers. TWS API documentation: Market Depth (Level II). interactivebrokers.github.io/tws-api/market_depth.html
- [6] Massive (formerly Polygon.io). Stocks WebSocket and REST API documentation. massive.com/docs
- [7] `ib_async`: Python framework for the Interactive Brokers API (successor to `ib_async`). github.com/ib-api-reloaded/ib_async
- [8] `yfinance`: Python library for Yahoo Finance market data.